

1.

N items are stored in a sorted doubly linked list. For a delete operation, a pointer is provided to the record to be deleted.

An algorithm performs the delete operation on the list. What will be the time complexity to perform this task.:

- A. $O(\log N)$
- B. $O(N)$
- C. $O(N^2)$
- D. $O(1)$

Answer: D

2.

The following code deletes the rightmost node of a doubly linked list. Identify the missing statement

```
right = right->prev;  
?  
right->next = NULL
```

- A. `free(right);`
- B. `free(right->next);`
- C. `free(right->prev);`
- D. none of these

Answer: B

3.

Match the following:**How many link fields get updated in doubly linked list?**

- W. Inserting a node right most(while considering that previous and next pointer of the new node are not null) 2**
- X. Deleting a node right most 4**
- Y. Inserting a node elsewhere in the middle 1**
- Z. Deleting a node elsewhere in the middle 3**

- A. W-3 X-4 Y-2 Z-1**
- B. W-3 X-1 Y-4 Z-2**
- C. W-4 X-4 Y-2 Z-1**
- D. W-3 X-4 Y-1 Z-2**

Answer: B

4.

The time required to search an element in a linked list of length n is

- A. $O(\log_2 n)$**
- B. $O(n)$**
- C. $O(1)$**
- D. $O(n^2)$**

Answer: B

5.

A doubly linked list is declared as :

struct node

```
{  
    struct node *prev;  
    int data;  
    struct node *next;  
};
```

Where next and backward link to the adjacent elements of the list. Which of the following segment code deletes the node pointed to by X from doubly linked list, if it is assumed that X points to neither the first nor the last node of the list.

- A. `X->prev->next=X->next; X->next->prev=X->prev`
- B. `X->prev->next = X->next; x->next->prev=X->prev;`
- C. `X->prev->next = X->prev; X->next->prev=x->prev;`
- D. `x->prev->next = x->prev; x->next->prev=x->next;`

Answer: B